DRA-PULSE: Localizing Persona Signals Across Deep Research Agent Pipelines

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Abstract

Deep Research Agents (DRAs) generate long-form reports through a multi-stage pipeline—planning, search, compression, and writing—conditioned on user personas. A key open question is *which* stage of this pipeline actually reflects persona conditioning, and how reliably. We introduce **N-way stage-wise persona attribution**: given a report artifact produced by one of N candidate personas, identify the ground-truth persona from intermediate pipeline outputs alone. Applying this framework to 120 reports generated by one DRA instantiation—Gemini 3.6 Flash via `open_deep_research` over PDR-Bench (20 query groups, 10 domains, two seeds)—we find a *dip-and-recovery* trajectory: attribution accuracy drops from Planning (0.808) to Search (0.550), remains statistically unresolved at Compression (0.592), and recovers at Writing (0.800), with macro Acc@1 of 0.688 against a 3-way chance level of 0.333. The Planning-to-Search drop is -0.258 (task-cluster bootstrap 95% CI $[-0.333, -0.175]$), while the Compression-to-Writing recovery is $+0.208$ ($[0.142, 0.283]$). At the report level, 32 cases lose correct attribution between Planning and Search, and 24 of these become attributable again by Writing. Compared with BM25 on the same candidate sets, Solar gains 0.175 at Planning and 0.225 at Writing, but not at Search or Compression. After conservatively masking named entities and candidate-derived identity phrases, the trajectory persists (0.725/0.558/0.575/0.783); masking reduces Planning by 0.083 but changes no later stage by more than 0.017. An independent GPT-5.4-nano replication on seed 0 yields the same shape (0.750/0.467/0.617/0.850), with 0.783 overall prediction agreement and Cohen's $\kappa = 0.763$ against Solar. We provide tracing, deterministic candidate ordering, non-LLM baselines, and analysis code as a reproducible diagnostic protocol.

1 Introduction

Persona conditioning is often evaluated only at the final answer, obscuring whether a research pipeline preserved the conditioning signal throughout its intermediate decisions. This matters for multi-stage agents: a final report may appear personalized even when planning or retrieval ignored the user, or it may recover information that was weakly expressed in an earlier artifact. We therefore treat stage artifacts as observable traces and ask how well an independent evaluator can recover the conditioning persona at each stage.

Our contributions are:

- We define the *N-way stage-wise attribution* task for DRA pipelines and decompose its aggregate trajectory into report-level loss-and-recovery transitions.
- We build DRATracer, a deterministic LangGraph v2 event listener that captures four stage artifacts from a single pipeline run without post-hoc reconstruction.
- We conduct a controlled attribution study on PDR-Bench with two generation seeds, hard-negative candidates, non-LLM baselines, quality sensitivity analyses, and task-cluster bootstrap uncertainty.
- Across 120 reports, we find a report-level *loss-and-recovery* phenomenon: Search falls 25.8 points below Planning, but 24 of the 32 reports that lose attribution at Search recover by Writing.
- We show that evaluator advantage is stage-specific: Solar clearly exceeds BM25 at Planning and Writing, while their Search and Compression differences are not distinguishable from zero. The trajectory also survives a conservative candidate-derived identity-masking control and replicates with an independent GPT-5.4-nano evaluator.

2 Related Work

2.1 Personalized Deep Research

Personalization in information retrieval has a long history of using user profiles and behavioral histories to re-rank search results (Teevan et al., 2005). Recent work has extended personalization to large language model (LLM) generation, with benchmarks such as LaMP (Salemi et al., 2023) testing whether models can condition outputs on user-specific history. **PDR-Bench** (Liang et al., 2025) is the first benchmark specifically targeting personalized *deep research*—multi-step, long-form report generation conditioned on rich user personas. PDR-Bench provides 25 personas and 250 English query pairs spanning ten domains, with personas encoding both identity attributes (demographics, occupation) and actionable preferences (information-seeking style, decision-making patterns).

A key gap in PDR-Bench and prior personalization benchmarks is the absence of a stage-level diagnostic: they measure whether a report *looks* personalized, but not where the conditioning persona remains recoverable inside the pipeline. We address this gap by framing personalization as an N-way attribution problem applied at each pipeline stage independently.

2.2 Authorship Attribution and Persona Identification

Classical authorship attribution identifies the author of a text from a closed candidate set, relying on stylometric features such as function-word distributions and syntactic patterns (Stamatatos, 2009). Neural approaches learn task-specific classifiers and transferable authorship representations from text (Shrestha et al., 2017; Rivera-Soto et al., 2021).

Li and Chaturvedi (2025) is the closest prior work to ours: they propose *comparative personalization* for multi-document summarization, identifying fine-grained preference differences by comparing a target user against other users—effectively a pairwise preference attribution task. Our work departs from this line in two fundamental ways: (1) **N-way simultaneous** attribution ($N = 3$), without decomposing into pairwise decisions; and (2) **stage-wise** attribution across a four-stage DRA pipeline, enabling *localization* of where in the pipeline the conditioning signal is expressed or lost.

2.3 Multi-Step Agent Pipelines and Preference Fidelity

Deep research agents such as `open_deep_research` (LangChain AI, 2025) implement multi-step LangGraph pipelines: a query is first elaborated into a research brief, then dispatched to parallel researcher subgraphs that search and compress evidence, before a final synthesis pass generates the report. This pipeline architecture means that a persona conditioning signal applied once at input must propagate through four qualitatively distinct transformation stages.

PrefDisco (Li et al., 2025) finds that preference accuracy degrades specifically in the reasoning and content-selection layers of multi-step pipelines, motivating a stage-wise evaluation rather than end-to-end accuracy alone. We operationalize this motivation with a tracing adapter that captures deterministic artifacts at each stage, enabling per-stage attribution without post-hoc reconstruction.

Stage	Artifact	Content
Planning	a_{plan}	Research brief: scoped topics, intended angle
Search	a_{search}	Queries + top-3 Serper organic results each
Compression	a_{compress}	Per-topic synthesis of retrieved evidence
Writing	a_{write}	Full long-form research report

Table 1: Four stage artifacts captured per DRA run.

3 Method

3.1 Task Formulation

Let P be a DRA pipeline and $\mathcal{C} = \{p_1, \ldots, p_N\}$ be a candidate persona set ($N = 3$ in our primary experiments). P is conditioned on one **ground-truth persona** $p^* \in \mathcal{C}$ and a research query q, and produces a sequence of **stage artifacts**:

$$\mathcal{A}(q, p^*) = \big(a_{\text{plan}},\ a_{\text{search}},\ a_{\text{compress}},\ a_{\text{write}}\big). \quad (1)$$

The **N-way stage-wise attribution** task is defined as: for each stage artifact $a \in \mathcal{A}$, identify p^* from $\mathcal{C}$:

$$\hat{p} = f\big(a,\ \mathcal{C}\big), \qquad \hat{p} \in \mathcal{C}. \quad (2)$$

Attribution accuracy at stage s is $\text{Acc}_s = \mathbb{E}[\mathbf{1}[\hat{p}_s = p^*]]$, with chance level $1/N$. Stage-wise accuracy profiles localize *where* in the pipeline the conditioning signal is expressed: a flat profile suggests the signal never propagates; a peak at a_{plan} that decays by a_{write} suggests early capture but downstream dilution.

3.2 DRA Pipeline and Artifact Schema

We use **open_deep_research** (LangChain AI, 2025) as the DRA backbone, instantiated with Gemini 3.6 Flash. The pipeline implements a four-stage LangGraph graph; Table 1 summarizes each stage's artifact.

DRATracer is our custom LangGraph v2 event listener that captures all four artifacts deterministically from a single pipeline run. The key implementation challenge is that three researcher subgraphs run concurrently under the supervisor; DRATracer resolves per-artifact topic identity using parent run-ID correlation across `on_chain_start` and `on_chain_end` events. All artifacts are serialized to a versioned JSON schema (`dra_trace_v2`) with SHA-256 prompt hashes and a query/source ledger for reconciliation.

3.3 Attribution Protocol

LLM Matcher. We use Upstage Solar Pro (API model ID: `solar-pro`) as the primary attribution model (temperature $= 0$, structured output). Given artifact text a and persona set $\mathcal{C}$ (order-shuffled to prevent position bias), the matcher returns a single predicted persona $\hat{p}$. Candidate order is deterministically shuffled

Condition	Manipulation	Tests
Full (baseline)	Unmodified persona	Upper bound
Identifier-masked	NER + all-candidate identity-phrase masking	Surface mentions?
Shuffled-actionable	Donor actionable prefs, shell identity	Actionable vs. identity?

Table 2: Shortcut control conditions executed in this paper. Identifier masking is a post-processing pass on all four existing stage artifacts; shuffled-actionable is a 15-report generation-time intervention. Two additional generation-time conditions (actionable-only, identity-only) were attempted but did not pass the pre-specified completeness gate (Appendix D).

per run and stage using a SHA-256-derived seed. To bound matcher context consistently, serialized artifacts are truncated to 4,000 characters for Planning, 3,500 for Search, 6,000 for Compression, and 8,000 for Writing; over-length Writing artifacts retain both the opening and tail of the report.

Non-LLM Baselines. We include three baselines: (1) *Random*, uniform over $\mathcal{C}$ (expected accuracy $= 1/N$); (2) *BM25*, Okapi BM25 similarity between artifact and persona text; (3) *Embedding*, cosine similarity via a sentence encoder.

Hard Negatives. For each ground-truth persona, the two distractor personas are the top-2 same-domain personas by actionable-token Jaccard similarity (≥ 0.2 = hard negative; otherwise nearest fallback, seeded per task). This ensures chance-level performance is a meaningful baseline. Candidate-set-size sensitivity uses prefixes of this same deterministic ranking: top-1 for $N = 2$ and top-4 for $N = 5$.

3.4 Shortcut Controls

High attribution accuracy could arise from **lexical leakage**—name tokens, demographic phrases, or verbatim persona text appearing in an artifact—rather than broader persona-conditioned content selection. We control for this with three generation-time conditions and one post-processing condition (Table 2).

4 Experimental Setup

4.1 Datasets

PDR-Bench (Primary). We use the English split of PDR-Bench (Liang et al., 2025), which provides 25 user personas and 250 query-persona pairs across ten domains (Education, Career, Health, Travel, Finance, Creative, Shopping, Real Estate, Law, and Parenting). We apply domain-stratified largest-remainder sampling to select 20 confirmatory query groups; eligibility filtering removes tasks with fewer than three personas that have at least one actionable token after identity masking. Hard-negative candidate sets are constructed as

Run	Groups	N	Seeds	Reports
PDR-Bench confirmatory (full)	20	3	2	120
Total				**120**

Table 3: Generation budget (excludes 15-report dev pilot).

described in §3. The confirmatory run uses two seeds ($s \in \{0, 1\}$), yielding **120 reports** (20 groups × 3 personas × 2 seeds). A 5-group development split (15 reports) serves as an engineering gate and is excluded from all research-question analyses.

4.2 Generation Stack

All reports are generated with open_deep_research (LangChain AI, 2025) using Gemini 3.6 Flash for all pipeline roles. Search is provided by a Serper adapter (max 7 queries per report, top-3 organic results each, max 15 unique URLs, global hard stop at 2,400 queries). Generation seed is fixed via `generation_config.seed`; other hyperparameters use model defaults. Every artifact records a full `execution_config` block including model IDs, seeds, and SHA-256 hashes for reproducibility. The primary analysis includes all schema-valid reports. We separately report a sensitivity analysis restricted to runs with no completeness errors and to runs satisfying all pre-defined success criteria.

4.3 Report Budget

Table 3 summarizes the generation runs analyzed in this paper. Identifier-masked variants are produced by post-processing existing artifacts at zero additional cost.

4.4 Evaluation Metrics

Attribution Accuracy (Acc@1). Fraction of reports for which the matcher's top-1 prediction equals p^*. Reported per stage and macro-averaged across seeds. Chance level: $1/N = 0.33$.

Stage Trajectory. The four-tuple $(\mathrm{Acc}_{\mathrm{plan}}, \mathrm{Acc}_{\mathrm{search}}, \mathrm{Acc}_{\mathrm{compress}}, \mathrm{Acc}_{\mathrm{write}})$ plotted over pipeline depth. A rising trajectory implies signal *accumulation*; a falling trajectory implies *early capture but downstream dilution*.

Seed Variance. Std. dev. of Acc@1 across seeds 0 and 1 as a fragility indicator.

5 Results

5.1 Stage-Wise Attribution Accuracy

Table 4 reports the controlled two-seed comparison between Solar Pro and three non-LLM baselines. Solar is strongest overall, but its advantage over BM25 is stage-dependent. The primary trajectory is also shown in Figure 1.

Stage	Solar	BM25	Embed	Rnd
Planning	**0.808**	0.633	0.442	0.292
Search	0.550	**0.567**	0.292	0.367
Compression	**0.592**	0.567	0.375	0.383
Writing	**0.800**	0.575	0.475	0.250
Macro avg.	**0.688**	0.585	0.396	0.323
Chance		0.333		

Table 4: N-way stage-wise attribution Acc@1 on PDR-Bench confirmatory split (seeds 0 and 1, $n = 120$, $N = 3$ hard-negative candidates). Solar = Solar Pro (LLM matcher); BM25 = Okapi BM25; Embed = all-MiniLM-L6-v2 cosine; Rnd = random.

Beyond lexical matching. The Solar–BM25 gap varies by stage. Across the paired two-seed reports, Solar exceeds BM25 by 0.175 at Planning (task-bootstrap 95% CI $[0.083, 0.258]$) and 0.225 at Writing ($[0.142, 0.308]$). At Search, BM25 is 0.017 higher than Solar, but the paired difference includes zero ($[-0.125, 0.092]$); the Compression difference is similarly unresolved ($+0.025$, $[-0.058, 0.117]$). Thus, the LLM evaluator's advantage is concentrated at Planning and Writing, while the low Search score is shared with a strong lexical baseline. Critically, both non-LLM baselines independently reproduce the Planning-to-Search dip: BM25 drops from 0.633 to 0.567, and Embedding from 0.442 to 0.292. Because these baselines involve no LLM reasoning, the dip pattern reflects artifact-level properties rather than an evaluator-specific bias.

Candidate-construction sensitivity. Because per-ground-truth hard negatives are selected around the target persona, the candidate set can contain a structural prior. An artifact-free heuristic that selects the candidate with the highest mean pairwise actionable-token Jaccard identifies 27 of 60 construction centers (0.450; task-cluster bootstrap 95% CI $[0.333, 0.567]$). We therefore treat absolute hard-negative Acc@1 with caution. Importantly, the same report keeps one candidate set across all stages, and the dip-and-recovery shape also appears under a symmetric task-shared cohort: 0.733/0.542/0.608/0.708. Candidate construction affects absolute levels but does not account for the replicated stage ordering.

Protocol-matched shortcut controls. We mask both spaCy PERSON/ORG/GPE/NORP/FAC entities and exact identity-field phrases derived from all three candidate profiles, including names, ages, occupations, education, residence, and family fields. Applying phrases from all candidates avoids using the ground-truth profile alone to define the transformation. The masked and full conditions have identical candidate sets and SHA-256 orders for all 480 paired decisions. Results are shown in Table 5.

The masked trajectory still drops from Planning to Search (-0.167, $[-0.258, -0.067]$) and recovers

Stage	Full	Masked	Δ
Planning	0.808	0.725	-0.083
Search	0.550	0.558	$+0.008$
Compression	0.592	0.575	-0.017
Writing	0.800	0.783	-0.017

Table 5: Solar Acc@1 before and after candidate-derived identity masking. Only Planning changes reliably: $\Delta = -0.083$, task-cluster bootstrap 95% CI $[-0.150, -0.025]$. Intervals for the other three deltas include zero.

Stage	Solar	GPT-nano	Agree	κ
Planning	0.800	0.750	0.850	0.837
Search	0.533	0.467	0.717	0.680
Compression	0.567	0.617	0.700	0.673
Writing	0.817	0.850	0.867	0.855
All	0.679	0.671	0.783	0.763

Table 6: Per-stage cross-matcher comparison (seed 0, $n = 60$). Solar = Solar Pro; GPT-nano = GPT-5.4-nano-2026-03-17. Agreement and κ are highest where attribution accuracy is highest (Planning, Writing) and lowest where it is lowest (Search, Compression).

from Compression to Writing ($+0.208$, $[0.100, 0.308]$). Thus, direct identity phrases explain part of Planning recoverability but do not explain the late-stage recovery.

Cross-matcher replication. On the frozen seed-0 cohort, GPT-5.4-nano produces 0.750/0.467/0.617/0.850 across Planning, Search, Compression, and Writing. Its Planning-to-Search drop is -0.283 (task-cluster bootstrap 95% CI $[-0.433, -0.133]$), followed by a Compression-to-Writing gain of $+0.233$ ($[0.083, 0.383]$). Solar and GPT agree on 188 of 240 decisions (0.783; Cohen's $\kappa = 0.763$). Stage-wise κ mirrors the primary trajectory: highest at Writing (0.855) and Planning (0.837), and lower at Search (0.680) and Compression (0.673)—the two stages with the lowest attribution accuracy. The replicated trajectory is therefore not specific to the primary evaluator. Table 6 gives per-stage detail.

Dip-and-recovery trajectory. Across both seeds, accuracy drops from Planning (0.808) to Search (0.550, $\Delta = -0.258$, 95% CI $[-0.333, -0.175]$), changes little at Compression (0.592, $\Delta = +0.042$, $[-0.025, 0.108]$), and recovers at Writing (0.800, $\Delta = +0.208$, $[0.142, 0.283]$). Planning and Writing are nearly equal in the aggregate. The trajectory appears in both generation seeds; the largest seed difference is 0.050 at Compression, and every seed-difference interval includes zero.

Run-level distribution. Of 120 reports, 55 (45.8%) are correctly attributed at all four stages, while 11 (9.2%) fail at every stage. The remaining 54 reports are

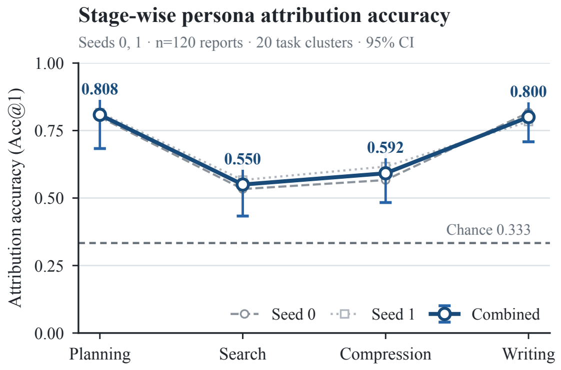

Figure 1: Two-seed stage-wise attribution trajectory ($n = 120$). Thick line: combined estimate; thin lines: individual seeds; error bars: 95% task-cluster bootstrap intervals.

Transition	Lost	Gained	Net
Planning → Search	32	1	−31
Search → Compression	8	13	+5
Compression → Writing	6	31	+25

Table 7: Report-level attribution-state transitions over 120 reports. Lost denotes correct→wrong; Gained denotes wrong→correct.

distributed across one to three correct stages, indicating both persistent report-level difficulty and stage-specific changes.

Report-level loss and recovery. Table 7 decomposes the aggregate curve into changes for the same reports. Planning→Search has 32 losses but only one gain. Compression produces 13 gains against eight losses, but its aggregate change remains statistically unresolved. Writing produces 31 gains against six losses. Of the 32 reports that are correct at Planning and wrong at Search, 11 recover by Compression and 24 (75.0%) recover by Writing. More broadly, 33 of the 54 Search errors (61.1%) are correct at Writing.

Quality sensitivity. Restricting analysis to the 90 reports satisfying all pre-defined success criteria yields Planning/Search/Compression/Writing accuracies of 0.822/0.567/0.611/0.811. Excluding the 29 reports with completeness errors yields 0.824/0.571/0.615/0.813 ($n = 91$). Both retain the same dip-and-recovery ordering as the all-report analysis.

6 Analysis

6.1 Why Does Search Accuracy Dip?

The Search artifact combines generated query strings with retrieved titles and snippets. Its attribution accuracy is the lowest stage value for Solar, and Planning-to-Search losses greatly outnumber gains (32 versus one). Table 8 shows Acc@1 broken down by artifact view and generation seed. A paired component control yields Acc@1 of 0.600 for queries alone and 0.533 for titles and snippets alone, versus 0.550 for the full Search artifact. Queries exceed full Search by 0.050 (task-cluster bootstrap 95% CI $[0.000, 0.108]$); this margin is identical in both generation seeds (+0.050 each). The direct queries-minus-snippets difference of 0.067 remains unresolved ($[-0.008, 0.142]$). Within-persona query strings share no lexical overlap across seeds (Jaccard $= 0.0$), confirming that query-view recoverability reflects semantic content rather than phrasing repetition. Cross-seed URL overlap within the same persona (Jaccard $= 0.042$) is only marginally higher than across personas on the same query (0.034; task-cluster bootstrap CI $[-0.001, 0.019]$), suggesting that retrieved document sets are largely insensitive to persona conditioning at the Search stage. Thus, retrieved text does not restore recoverability and may dilute a modest query-carried signal, but neither view alone explains the full Planning-to-Search drop. Because these are post-hoc views, they do not identify a causal retrieval effect.

View	Seed 0	Seed 1	Combined
Full Search	0.533	0.567	0.550
Queries only	0.583	0.617	0.600
Snippets only	0.483	0.583	0.533

Table 8: Search Acc@1 by artifact view and generation seed ($n = 60$ per seed; Solar Pro matcher; 240 total decisions, SHA-256 order audit 240/240 pass). The queries-only advantage over full Search is +0.050 in each seed independently.

Compression's point estimate rises only 0.042 and its interval includes zero. Writing then adds 0.208, with 31 report-level gains against six losses. Solar's larger advantage over BM25 at Writing is consistent with later artifacts containing evidence beyond direct lexical overlap, but it does not establish which linguistic or content-selection mechanism produces that evidence. The results therefore localize recoverability changes without establishing causal signal loss, harmful retrieval, or report utility.

Structural origin of Writing recovery. The Writing stage prompt in open_deep_research re-injects the *research brief*—the Planning artifact—as its top-level instruction (the template includes <Research Brief>{research_brief}</Research Brief> before the compressed findings). Because the research brief is itself persona-conditioned, this creates an architectural path by which the persona signal suppressed at Search reappears in Writing: the Writing model reads the persona-aligned planning scope again before synthesising retrieved evidence. This hypothesis is consistent with three observations. First, Writing

GT	Predicted	Stage	Count
User18	User10	Search	6
User12	User10	Search	4
User12	User10	Compression	4
User17	User10	Search	4
User18	User10	Compression	4
User18	User10	Writing	4
User18	User8	Search	4

Table 9: Persona confusion pairs with ≥ 4 stage-level errors across 120 reports. User10 concentrates misattributions at Search and Compression, the two lowest-accuracy stages.

accuracy (0.800) closely matches Planning (0.808), as expected if the same source of signal is active at both stages. Second, Writing accuracy under identity masking remains 0.783—only 0.017 below the masked Planning accuracy (0.725)—whereas Search and Compression show no reliable reduction; this is consistent with the persona signal surviving as semantic content rather than surface identity phrases. Third, BM25 also recovers slightly at Writing (0.575) relative to Search and Compression (both 0.567), consistent with lexical persona content being re-emphasised via the research brief. We cannot rule out that additional mechanisms, such as the model's conditioning temperature or the length of the Writing artifact, contribute to the recovery; disentangling these factors is a direction for future work.

6.2 Persona Confusability

Not all persona pairs are equally distinguishable. While errors span multiple pairs, one persona (User10) appears as the predicted label across several ground-truth identities at Search and Compression—including User18→User10 (six times at Search, the single largest confusion), User12→User10, and User17→User10. Table 9 lists all ground-truth/prediction pairs with four or more stage-level errors. This pattern suggests User10 may act as a generic attractor at the lower-recoverability stages, consistent with high pairwise actionable-token similarity among hard-negative candidates. A systematic analysis of overlap and pair-level difficulty remains future work.

6.3 Shortcut Controls

Identity masking. The strong Planning reduction under masking indicates that early-stage recoverability partly uses direct identity evidence. In contrast, Search, Compression, and Writing change by at most 0.017, and the masked Writing accuracy remains 0.783. This is consistent with later recoverability drawing on evidence beyond exact candidate identity phrases. It does not prove that the remaining evidence corresponds exclusively to actionable preferences: paraphrased identity cues and other lexical shortcuts may remain.

Stage	Shell	Donor	Other	Δ	95% CI
Planning	0.667	0.133	0.200	-0.533	$[-0.933, -0.133]$
Search	0.467	0.333	0.200	-0.133	$[-0.600, +0.200]$
Compression	0.533	0.200	0.267	-0.333	$[-0.733, +0.000]$
Writing	0.800	0.133	0.067	-0.667	$[-0.933, -0.400]$

Table 10: Shuffled-actionable intervention: Solar prediction rates for the identity shell, the actionable donor, and the third candidate ($n = 15$, seed 0). Δ = Donor−Shell; 95% CI: task-cluster bootstrap. Search is the only stage where donor and shell are not clearly separated.

Shuffled-actionable intervention. In a frozen 15-report seed-0 subset, we replace each identity shell's actionable preferences with those of another candidate persona. Table 10 shows prediction rates for the shell, the actionable donor, and the third candidate at each stage. Predictions strongly favor the shell over the donor at Planning (0.667 vs. 0.133; donor-minus-shell -0.533, $[-0.933, -0.133]$) and Writing (0.800 vs. 0.133; -0.667, $[-0.933, -0.400]$). Search is the only stage where donor and shell prediction rates approach parity (-0.133, $[-0.600, +0.200]$), consistent with its role as the lowest-recoverability stage. The Writing result holds in the strict-quality subset ($n = 8$: shell 0.875, donor 0.125). Thus, detectable output identity remains anchored more strongly to the identity shell than to transplanted actionable preferences; this does not establish that the model ignores actionable preferences.

7 Limitations

Attribution measures whether a matcher can recover the conditioning persona; it does not measure usefulness, factual quality, user satisfaction, or causal influence. Our primary result uses one DRA backbone (Gemini 3.6 Flash via `open_deep_research`); whether the dip-and-recovery pattern generalises to other architectures, LLM families, or pipeline designs remains open. We also use a primary LLM matcher with one seed replicated by a second matcher, a three-candidate closed set, and one benchmark; neither matcher has been validated against human attributors. The two generation seeds share the same 20 task clusters, so uncertainty intervals cluster by task but cannot capture broader dataset or model variation. Candidate-derived identity masking is conservative: NER false positives can remove task-relevant content, while paraphrased identity cues may remain. The per-ground-truth hard-negative construction also provides a weak candidate-only structural prior; we report a symmetric task-shared sensitivity because this can affect absolute accuracy. Our Search views are post-hoc, while the shuffled-actionable intervention is small (15 reports, one seed; eight strict-quality reports). Neither establishes downstream causal effects or utility. GPT replication covers seed 0 only, and cross-dataset replication remains an important next step.

8 Conclusion

We introduced N-way stage-wise persona attribution as a diagnostic for Deep Research Agent pipelines and showed that persona conditioning leaves measurable, stage-specific traces across planning, search, compression, and writing artifacts. The primary finding is a report-level *loss-and-recovery* process rather than merely a non-monotonic average: 32 reports lose attribution from Planning to Search, and 24 of them recover by Writing. Search has the lowest persona recoverability (Acc@1 0.550); Compression remains statistically unresolved at 0.592, while Writing recovers to 0.800, nearly matching Planning at 0.808. The Solar–BM25 advantage is clear at Planning and Writing, but not at Search or Compression. The same shape remains after conservative identity masking (0.725/0.558/0.575/0.783) and under a symmetric task-shared candidate protocol. Together, these results localize a robust early-stage drop and late-stage recovery without establishing causal signal loss, harmful retrieval, or downstream utility. A generation-time shuffled-actionable control further favors the identity shell over the actionable donor at Writing (0.800 vs. 0.133), exposing a gap between identity recoverability and actionable-preference transmission. GPT-5.4-nano independently reproduces the trajectory and agrees with Solar on 78.3% of decisions ($\kappa = 0.763$), reducing dependence on a single evaluator.

Future work includes scaling the candidate set size ($N > 3$), extending to LaMP-QA for domain transfer, and developing persona-aware query diversification strategies to improve signal retention at the Search stage.

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A Full Quality Sensitivity

Table 11 reports Solar Acc@1 under four inclusion policies. The dip-and-recovery ordering is preserved under every restriction, confirming that the primary result is not driven by low-quality reports.

Policy	Plan	Search	Comp.	Write	n
All matched	0.808	0.550	0.592	0.800	120
Success criteria met	0.822	0.567	0.611	0.811	90
No completeness err.	0.824	0.571	0.615	0.813	91
No ledger errors	0.807	0.546	0.588	0.798	119

Table 11: Solar Acc@1 under four quality-filtering policies (Solar Pro matcher, per-GT hard-negative candidates). All policies preserve the same stage ordering.

B Per-Seed Baseline Results

Table 12 disaggregates the non-LLM baseline accuracies by generation seed. BM25 slightly exceeds Solar at Search in both seeds, confirming that the lexical-baseline parity at Search is not seed-specific. Random accuracy varies around the 0.333 chance level, as expected.

Seed	Method	Plan	Search	Comp.	Write
0	BM25	0.667	0.600	0.583	0.583
0	Embedding	0.450	0.300	0.383	0.483
0	Random	0.350	0.367	0.400	0.250
1	BM25	0.600	0.533	0.550	0.567
1	Embedding	0.433	0.283	0.367	0.467
1	Random	0.233	0.367	0.367	0.250

Table 12: Non-LLM baseline Acc@1 by generation seed ($n = 60$ per seed; per-GT hard-negative candidates). BM25 leads at Search in both seeds.

C Extended Persona Confusion Table

Table 13 lists all ground-truth/prediction pairs with two or more stage-level errors across the 120 reports. User10 appears as the predicted label for eight distinct ground-truth personas, concentrated at Search and Compression.

D EXP-011: Generation-time Ablation Outcomes

We attempted seed-0 generation for two generation-time ablation conditions: *actionable-only* (identity leaf keys removed from the persona conditioning prompt) and *identity-only* (only identity leaf keys retained). Each condition targeted 15 reports (5 task clusters $\times$ 3 personas, seed 0).

Both conditions produced 15/15 schema-valid artifacts. However, each violated the pre-specified completeness gate ($\leq$ 2 reports with missing topic sources), with 3 completeness issues per condition. The actionable-only condition additionally had one query-source ledger mismatch. Consequently, `generation_gate_passed = false` for both conditions, and seed-1 generation was not approved under the pre-registered gate protocol (DEC-006).

The Solar matcher was not run on these artifacts. Because the gate failure disqualifies these results from confirmatory use, we do not report attribution accuracy for the ablation conditions. The primary shortcut evidence in this paper relies instead on the post-hoc identifier-masking control (Table 5) and the generation-time shuffled-actionable intervention (Table 10), both of which passed their pre-specified quality criteria.

E Matcher Prompt Template

The LLM matcher receives a fixed system prompt and a stage-specific user prompt constructed as follows.

System prompt.

> You are an expert document analyst specializing in user profiling. Given a document excerpt from a research report and a set of candidate user profiles, identify which user this document was most likely produced for. Focus on content priorities, framing, vocabulary, and topic angles that reflect the user's background, goals, and preferences. Call `submit_attribution` with your answer.

GT	Predicted	Stage	Count
User18	User10	Search	6
User12	User10	Compression	4
User12	User10	Search	4
User17	User10	Search	4
User18	User10	Compression	4
User18	User10	Writing	4
User18	User8	Search	4
User10	User18	Compression	3
User11	User10	Search	3
User14	User13	Search	3
User14	User20	Search	3
User17	User10	Compression	3
User17	User10	Planning	3
User17	User10	Writing	3
User18	User8	Compression	3
User19	User10	Compression	3
User19	User10	Writing	3
User10	User8	Search	2
User11	User10	Compression	2
User12	User10	Planning	2
User13	User10	Search	2
User13	User14	Compression	2
User13	User14	Planning	2
User13	User14	Search	2
User13	User14	Writing	2
User14	User13	Compression	2
User17	User25	Compression	2
User17	User25	Search	2
User17	User25	Writing	2
User18	User10	Planning	2
User19	User10	Search	2
User19	User18	Planning	2
User19	User18	Search	2
User2	User1	Compression	2
User22	User10	Search	2
User5	User2	Search	2
User8	User10	Writing	2
User8	User18	Search	2
User9	User11	Compression	2
User9	User12	Compression	2
User9	User12	Search	2

Table 13: All persona confusion pairs with ≥ 2 stage-level errors across 120 reports (Solar Pro matcher). User10 concentrates misattributions across eight ground-truth personas at Search and Compression.

User prompt structure.

1. A stage-specific header (*Research Planning Brief*, *Search Queries and Retrieved Sources*, *Compressed Research Summaries*, or *Final Research Report (excerpt)*) followed by the artifact text, truncated to the stage character limit (Planning 4,000; Search 3,500; Compression 6,000; Writing 8,000 with opening and tail preserved).

2. A **Candidate User Profiles** section listing each candidate as **Candidate A / B / C** with the full persona text.

3. A closing instruction: *"Which candidate profile (A, B, or C) best matches the perspective, priorities, and content framing of the document above? Call* `submit_attribution`*."*

Candidate order is deterministically shuffled per run and stage using a SHA-256-derived seed, ensuring position bias cannot produce a systematic accuracy advantage for any fixed candidate slot. The matcher is called with temperature 0 and a structured `submit_attribution` tool that enforces a single-label response.

F DRATracer Artifact Schema

DRATracer is a LangGraph v2 event listener that captures all four pipeline artifacts deterministically from a single pipeline run and writes them as a `dra_trace_v2` JSON object. Table 14 summarises the top-level fields.

Field	Contents
schema_version	Schema identifier (dra_trace_v2)
run_id	Unique run identifier
research_brief	Planning stage artifact (text)
research_topics	Ordered topic list from the planning stage
search_trace	List of per-topic search records: queries issued, sources selected with titles and snippets
compressed_research	Per-topic evidence synthesis (Compression stage)
final_report	Full Writing stage artifact (text)
execution_config	Generation seed, model IDs, prompt SHA-256, manifest version, task/persona IDs
token_ledger	Input/output/total tokens, query counts (attempted, successful, failed), source counts
capture_completeness	Boolean flags for each required artifact field

Table 14: Top-level fields of the dra_trace_v2 artifact schema produced by DRATracer.

G Candidate Construction Protocol

For each ground-truth persona p^* in a task cluster, we select two distractor personas from the same domain as follows.

1. **Actionable-token extraction.** Each persona profile is tokenised (lowercased, whitespace-split) and identity-field tokens (name, age, occupation, education, residence, family) are removed, leaving the *actionable token set*.

2. **Jaccard similarity.** For each candidate persona $p_i \neq p^*$ in the same domain, we compute the Jaccard similarity of the actionable token sets: $J(p^*, p_i) = |A^* \cap A_i| / |A^* \cup A_i|$.

3. **Hard-negative selection.** The two personas with the highest $J(p^*, \cdot)$ among those satisfying $J \geq 0.2$ are chosen as distractors. If fewer than two satisfy this threshold, the nearest remaining same-domain personas are used as fallback (tie-broken by ascending user ID).

4. **Frozen per-GT set.** Each ground-truth persona has its own two-distractor set. The resulting $N = 3$ candidate set is fixed in the manifest and shared across all four stages and both generation seeds for that persona. Candidate presentation order is independently shuffled per run and stage using a SHA-256-derived seed.

An artifact-free heuristic that picks the candidate with the highest mean pairwise actionable Jaccard scores 0.450 (bootstrap 95% CI $[0.333, 0.567]$), establishing a structural prior above the 0.333 chance level. This motivates our additional task-shared sensitivity analysis (Section 5).